

Lecture: Above Threshold and Privacy Amplification by Subsampling

Consider the setting where we obtain queries in an online fashion; our interest is in returning those queries that are "large", privately. In the event that the queries were all known in advance, one can simply apply the exponential mechanism [Recall!] that samples from a suitably defined Gibbs distribution with error at most $c \cdot \frac{\log K}{\epsilon}$, when K is the number of queries, [why? what is the score function $q(\cdot, \cdot)$ here?] in order to identify that query that is the largest.

In the online setting, however, we shall confine our attention to identifying those queries that exceed some threshold $T \in \mathbb{R}$. In particular, we shall identify the "first" query that exceeds T .

The "Above Threshold" mechanism

Suppose that the data owner is fed a stream of queries $f_1, f_2, \dots$ that map $\mathcal{D}_N$ to $\mathbb{R}$. We assume for simplicity that $\Delta_{f_i} = 1$, $\forall i \geq 1$. We are interested in halting at and releasing the first query output that exceeds some fixed $T \in \mathbb{R}$, privately.

AboveThreshold

- Fix $\hat{T} = T + Z_1$, where $Z_1 \sim \text{Lap}(2/\epsilon)$.

- For each $i \geq 1$:

- Let $\bar{Z}_i \sim \text{Lap}(4/\epsilon)$.

- If $f_i(\mathcal{D}) + \bar{Z}_i \geq \hat{T}$:

 - * Output "Yes" and HALT.

- Else:

 - * Output "No" and CONTINUE.

Thm 1: AboveThreshold is ϵ -DP.

Proof: Fix databases $\underline{x} \sim \underline{x}'$. Let $\underline{a} = N^{t-1} Y$ be a sequence of possible outputs, for some $t \geq 1$ ($N \equiv$ "No" and $Y \equiv$ "Yes").

In what follows, we condition on $\bar{Z}^{t-1} = \bar{Z}^{t-1}$ and analyze the ratio of the probabilities that mechanism $M = \text{AboveThreshold}$ outputs $\underline{a}$ on $\underline{x}$ and $\underline{x}'$, conditioned on this event.

Let $g(\underline{x}) = \max_{i \leq t-1} (f_i(\underline{x}) + \bar{Z}_i)$. We hence have

$$\begin{aligned}
 P_h[M(\underline{x}) = \underline{a} \mid \bar{Z}^{t-1} = \bar{z}^{t-1}] &= P_h[\hat{\tau} > g(\underline{x}) \text{ and } f_t(\underline{x}) + \bar{Z}_t > \hat{\tau} \mid \bar{z}^{t-1}] \\
 &\stackrel{\text{constant, conditioned on } \bar{z}^{t-1}}{=} P_h[g(\underline{x}) \leq \hat{\tau} \leq f_t(\underline{x}) + \bar{Z}_t \mid \bar{z}^{t-1}] \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\bar{Z}_t}(z) \cdot f_{\hat{\tau}}(\tau) \cdot 1\{g(\underline{x}) \leq \tau \leq f_t(\underline{x}) + z\} dz d\tau \\
 &\quad \text{---(1)}
 \end{aligned}$$

Now, let $\hat{\tau} \triangleq \tau - g(\underline{x}) + g(\underline{x}')$ and $\hat{z} \triangleq z - g(\underline{x}) + g(\underline{x}') + f_t(\underline{x}) - f_t(\underline{x}')$.
 Clearly, we have $|\hat{\tau} - \tau| \leq 1$ and $|\hat{z} - z| \leq 2$. [Why?]

Plugging the change of variables back in (1), we see that

$$P_h[M(\underline{x}) = \underline{a} \mid \bar{z}^{t-1}] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\bar{Z}_t}(z) f_{\hat{\tau}}(\tau) \cdot 1\{g(\underline{x}) \leq \hat{\tau} \leq \hat{z} + f_t(\underline{x}')\} dz d\tau.$$

---(2)

Observe also that $\frac{f_{\bar{Z}_t}(z)}{f_{\bar{Z}_t}(\hat{z})} \leq \exp(\varepsilon/2)$ and $\frac{f_{\hat{\tau}}(\tau)}{f_{\hat{\tau}}(\hat{\tau})} \leq \exp(\varepsilon/2)$,

from the structure of the pdf of the Laplace distribution. Hence,

$$\begin{aligned}
 P_h[M(\underline{x}) = \underline{a} \mid \bar{z}^{t-1}] &\leq \exp(\varepsilon) \cdot \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\bar{Z}_t}(\hat{z}) f_{\hat{\tau}}(\hat{\tau}) \cdot 1\{g(\underline{x}') \leq \hat{\tau} \leq \hat{z} + f_t(\underline{x}')\} d\hat{z} d\hat{\tau} \\
 &= \exp(\varepsilon) \cdot P_h[M(\underline{x}') = \underline{a} \mid \bar{z}^{t-1}].
 \end{aligned}$$

Averaging over the values $\bar{z}^{t-1}$ completes the proof. $\square$

We now discuss the "utility" of AboveThreshold.

Def 1: AboveThreshold is said to be (α, β) -accurate for a sequence of K queries $f_1, \dots, f_K$, if, except with probability at most β , if AboveThreshold HALTs, then $f_t(x) \geq T - \alpha$, and if AboveThreshold returns "No", then $f_t(x) \leq T + \alpha$.

Theorem 2: For any sequences of K queries $f_1, \dots, f_K$ s.t. $f_i(x) < T - \alpha$, for all $i \leq K-1$, and $f_K(x) \geq T - \alpha$, we have that AboveThreshold is (α, β) -accurate, for $\alpha = \frac{8}{\epsilon} (\log K + \log(2/\beta))$.

Proof: We show that

$$\max_{i \in [K]} |\bar{z}_i| + |T - \hat{T}| \leq \alpha \quad \text{w.p.} \geq 1 - \beta. \quad (3)$$

If this is the case, we have that if $a_i = \text{"Yes"}$, for some $i \in [K]$,

$$\text{we must have } f_i(x) + \bar{z}_i \geq T + (\hat{T} - T)$$

$$\equiv f_i(x) \geq T + (\hat{T} - T) - \bar{z}_i \geq T - \alpha.$$

Likewise, if $a_i = \text{"No"}$, for some $i \in [K]$, we must have

$$f_i(x) + \bar{z}_i \leq T + (\hat{T} - T) \equiv f_i(x) \leq T + (\hat{T} - T) - \bar{z}_i \leq T + \alpha.$$

Now, note that $P[|\hat{T} - T| \geq \alpha/2] \leq \exp(-\frac{\epsilon \alpha}{4})$ and

$$P\left[\max_{i=1}^K |\bar{Z}_i| \geq \alpha/2\right] \leq K \cdot \exp\left(-\frac{\epsilon \alpha}{8}\right).$$

Setting the RHSs of the quantities above to be $\leq \beta/2$ gives rise to (3),
for $\alpha \geq \frac{8}{\epsilon} (\log K + \log(2/\beta))$. $\square$

Privacy amplification by subsampling

In this section, we take a detour from our exploration of specialized DP mechanisms and discuss a useful technique that is employed in machine learning applications [To be discussed!] for improving privacy parameters.

We consider the setting of a dataset $\underline{x}$ that is usually fed into an (ϵ, δ) -DP mechanism M ($\delta > 0$) for releasing a private estimate of a statistic. However, before we do so, we shall now "pre-process" $\underline{x}$ by **Poisson subsampling**: i.e., we shall pick each user i in $\underline{x}$ independently w.p. $p \in (0, 1)$ and collect the chosen records into a "subsampled dataset" $\underline{x}_p$ that is now given as input to M .

[In what follows, we assume that the (ϵ, δ) -DP guarantee of M is in the "add/remove" model.]

Theorem 3: The mechanism M on the subsampled dataset $\underline{x}_p$ obeys (ϵ', δ') -DP, where $\epsilon' = \log(1 + p(e^\epsilon - 1))$ and $\delta' = p\delta$.

Proof: Let $\underline{x} \in \mathbb{D}_N$ and $\underline{x}' = \underline{x} \setminus \{x_i\}$. We then have

$$\Pr[M(\underline{x}_p) = y] = (1-p) \Pr[M(\underline{x}_p) = y \mid x_i \notin \underline{x}_p] + p \cdot \Pr[M(\underline{x}_p) = y \mid x_i \in \underline{x}_p]$$

$$= (1-p) \Pr[M(\underline{x}'_p) = y] + p \cdot \Pr[M(\underline{x}'_p \cup \{x_i\}) = y]$$

$$\leq (1-p) \Pr[M(\underline{x}'_p) = y] + p \cdot (e^\epsilon \Pr[M(\underline{x}'_p) = y] + \delta)$$

$$= (1-p + pe^\epsilon) \cdot \Pr[M(\underline{x}'_p) = y] + p\delta.$$

Furthermore, by III^a arguments,

$$\Pr[M(\underline{x}_p) = y] \geq (1-p + pe^{-\epsilon}) \cdot \Pr[M(\underline{x}'_p) = y] - pe^\epsilon \delta$$

and hence

$$\Pr[M(\underline{x}'_p) = y] \leq \frac{\Pr[M(\underline{x}_p) = y] + pe^\epsilon \delta}{1-p + pe^{-\epsilon}}$$

$$\leq (1-p+pe^\epsilon) \cdot h[M(x_p)=y] + p\delta. \quad \square$$

↓
HW: Prove this inequality.